# Data Modeling in Power BI

**Data Modeling** in Power BI is the process of **organizing and structuring your data** so that reports and dashboards can be created efficiently. It ensures that **tables relate correctly**, and calculations are accurate.

## Key Concepts in Power BI Data Modeling

1. **Tables**
    - Imported datasets (Excel, SQL, CSV, etc.)
    - Can be fact tables or dimension tables (explained below)
2. **Relationships**
    - Define how tables are connected.
    - Types:
        - **One-to-Many (1:N)** → Most common
        - **Many-to-Many (N:N)** → Rare, use with care
    - Use primary key → foreign key mapping
3. **Calculated Columns**
    - New columns derived from existing columns using DAX formulas.
    - Example: `Profit = Revenue - Cost`
4. **Measures**
    - Aggregations computed at query time using DAX.
    - Example: `Total Sales = SUM(Sales[Amount])`
5. **Hierarchies**
    - Organize data in levels for drilling down.
    - Example: `Year → Quarter → Month → Day`
6. **Star & Snowflake Schema**

- Structuring tables for efficient analysis.

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# Star Schema

**Star Schema** is a simple data modeling approach where a **central fact table** connects directly to **dimension tables**.

## Characteristics

- Fact table in the center
- Dimension tables around it (like a star)
- Denormalized dimension tables (i.e., fewer joins)
- Optimized for **query speed and reporting**

## Structure

```
         Product
            |
Customer --- Sales --- Time
            |
         Store
```

## Example

- **Fact Table:** Sales
  - Columns: SalesID, ProductID, CustomerID, StoreID, Quantity, Revenue
- **Dimension Tables:**
  - Product → ProductID, Name, Category
  - Customer → CustomerID, Name, Region
  - Store → StoreID, Name, Location
  - Time → Date, Month, Quarter, Year

✅ **Pros:**

- Simple to understand
- Faster queries

- Easier for reporting and Power BI visuals

❌ **Cons:**

- Dimension tables can be large (denormalized)

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## 3️⃣ Snowflake Schema

**Snowflake Schema** is a more **normalized version** of the star schema. Dimension tables are **split into multiple related tables**.

### Characteristics

- Fact table in the center
- Dimension tables may have **sub-dimensions**
- More joins required
- Optimized for **storage efficiency**

### Structure

```
      Product
        |
    Product Category
        |
Customer --- Sales --- Time
        |
     Store
       |
   Store Region
```

### Example

- **Fact Table:** Sales
  - Same as star schema
- **Dimension Tables:**
  - Product → ProductID, Name, CategoryID
  - ProductCategory → CategoryID, CategoryName
  - Customer → CustomerID, Name, RegionID
  - Region → RegionID, RegionName

✅ **Pros:**

- Reduces data redundancy

- Saves storage

❌ **Cons:**

- More complex
- Slower query performance due to multiple joins

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